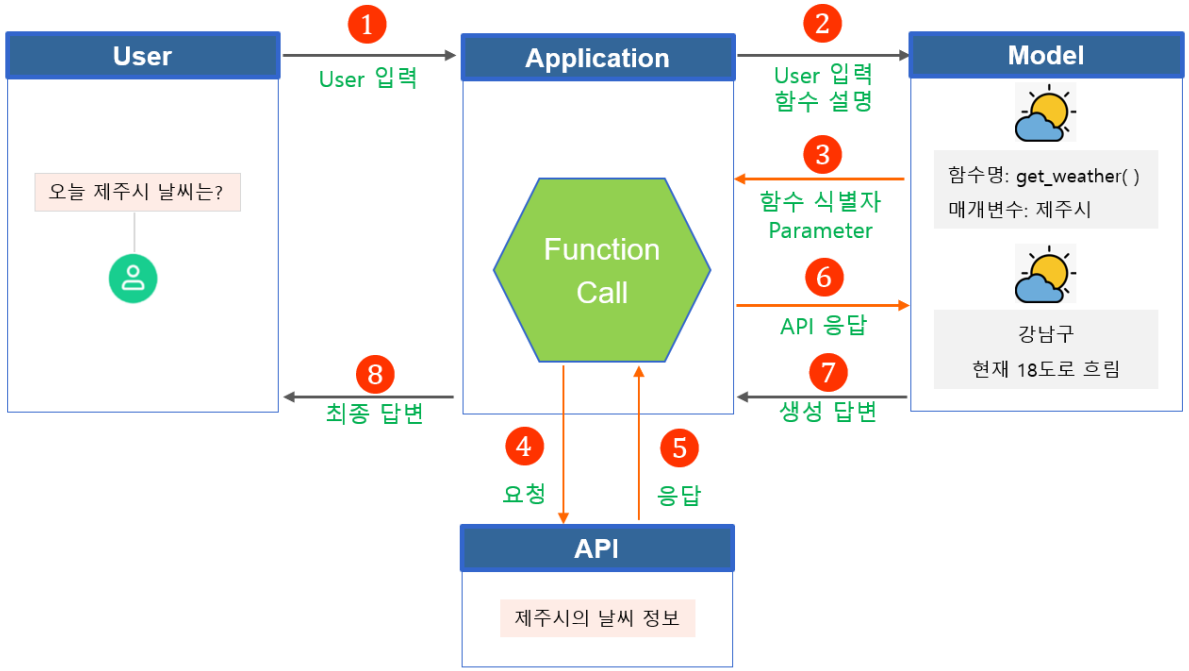
python + colab | OpenAI ChatGPT 4.0 | LangChain | LangGraph | AI Agent

Function calling



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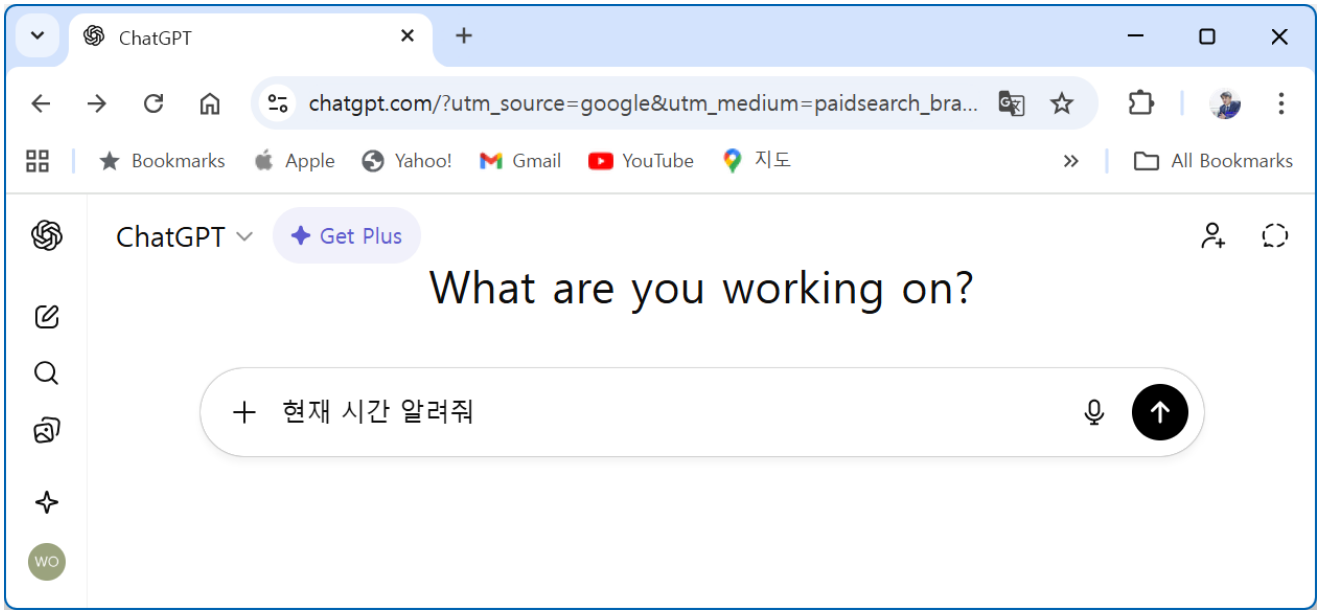
lesson Objectives

Contents

- What is a function calling?
- How Function Call Woks ?
  - Case I : Shopping LLM Agent
  - Case II : Task Automation Agent
  - Case III : Extract Information Agent
  - Practice I : Function Call
  - Practice II : Timezone Function Call
  - Practice III : Multi-Timezone Function Call
  - Practice IV : Function Call in Streamlit

What is a function calling? (1)



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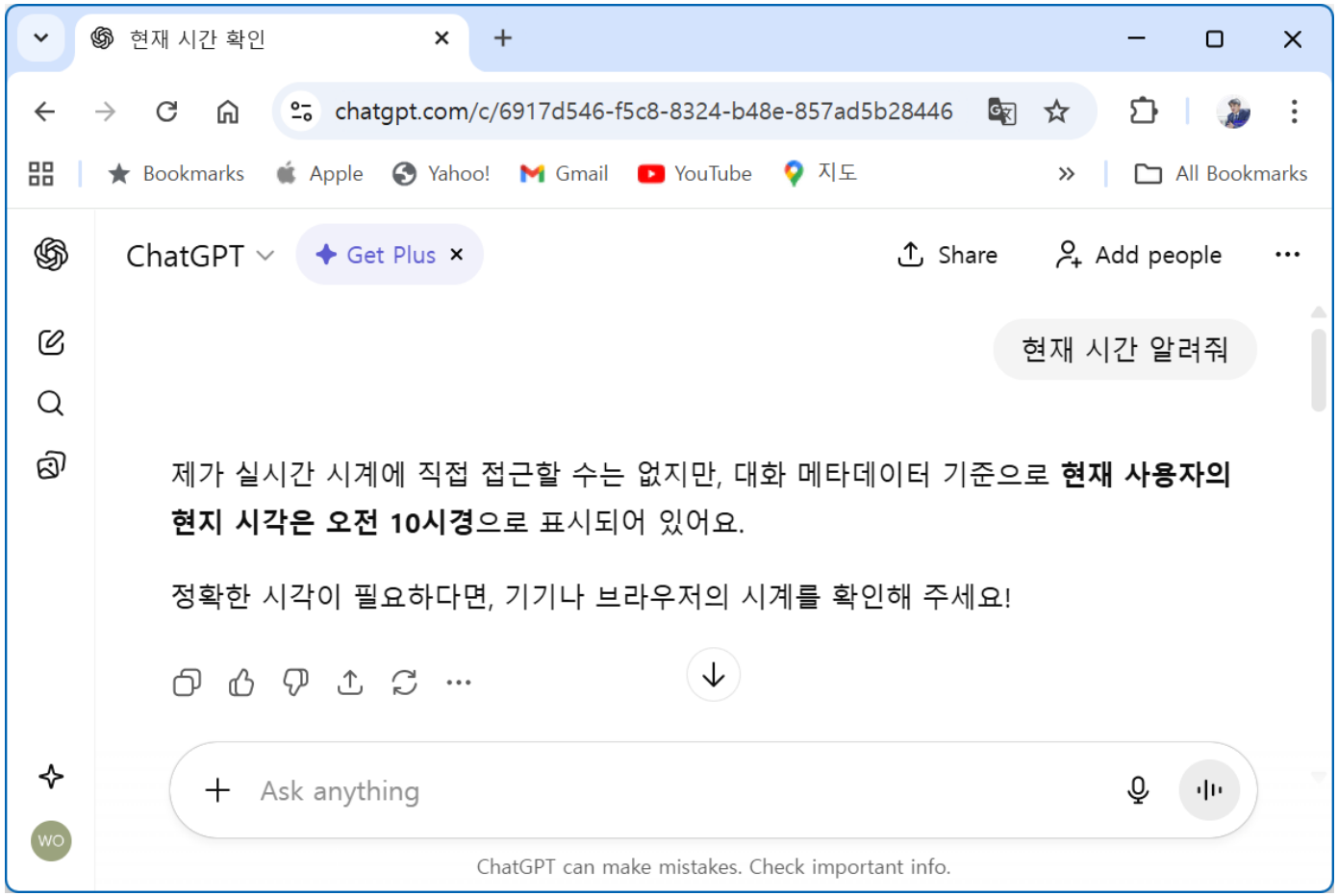
What is a function calling? (2)

Function calling

- Not support



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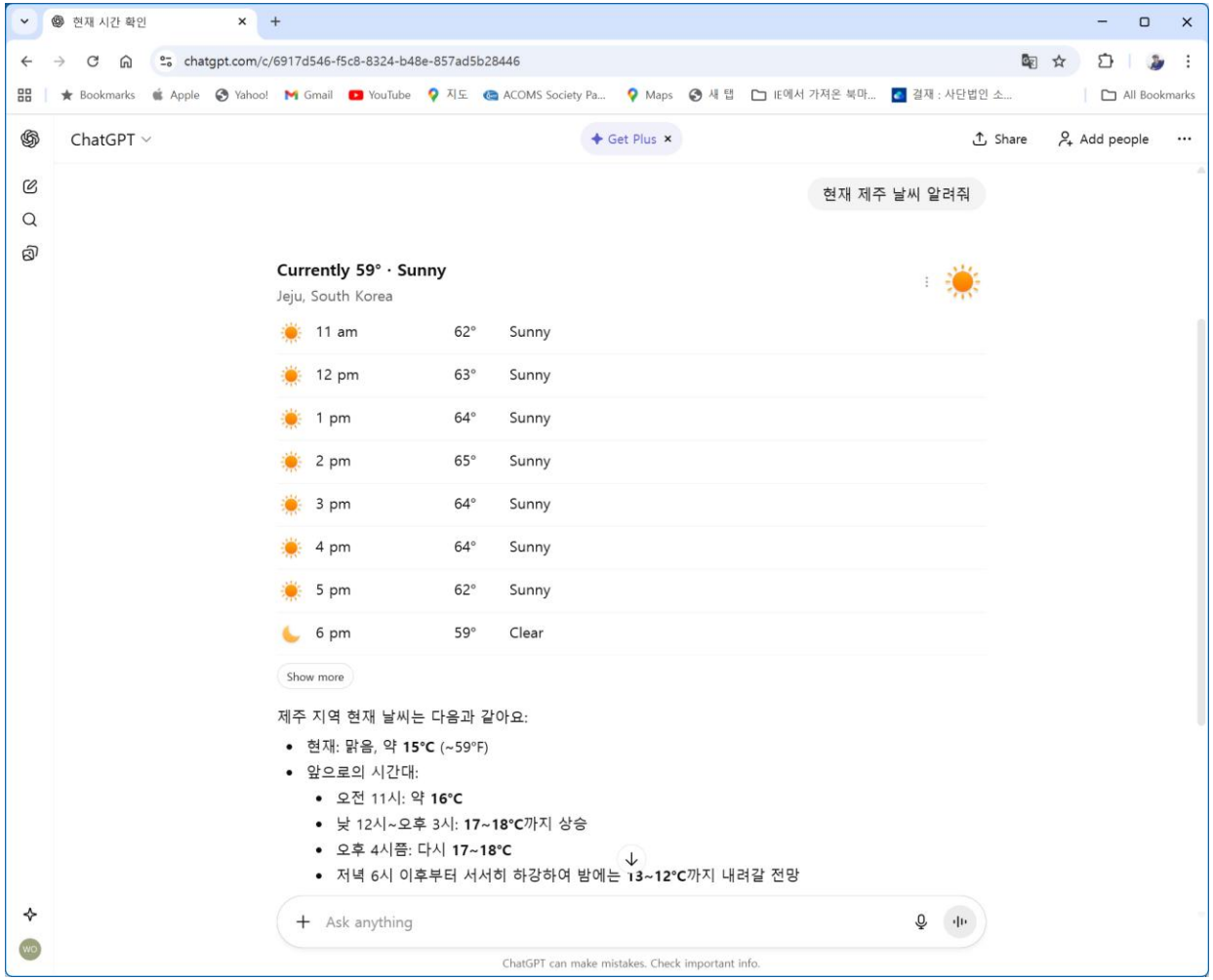
What is a Function Calling? (3)

Function calling

- Support



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What is a function calling? (4)

- Definition
  - Function calling **allows an AI model to interact with external tools** by generating a structured JSON output that specifies a function and its arguments to be executed by a developer's code.
  - Call APIs & functions within scripts or libraries.
  - The technology that emerged to overcome the limitations of LLM
    - Limitations of LLM
      - LLM answers based on pre-trained data
      - limitations in providing real-time information or utilizing external data.

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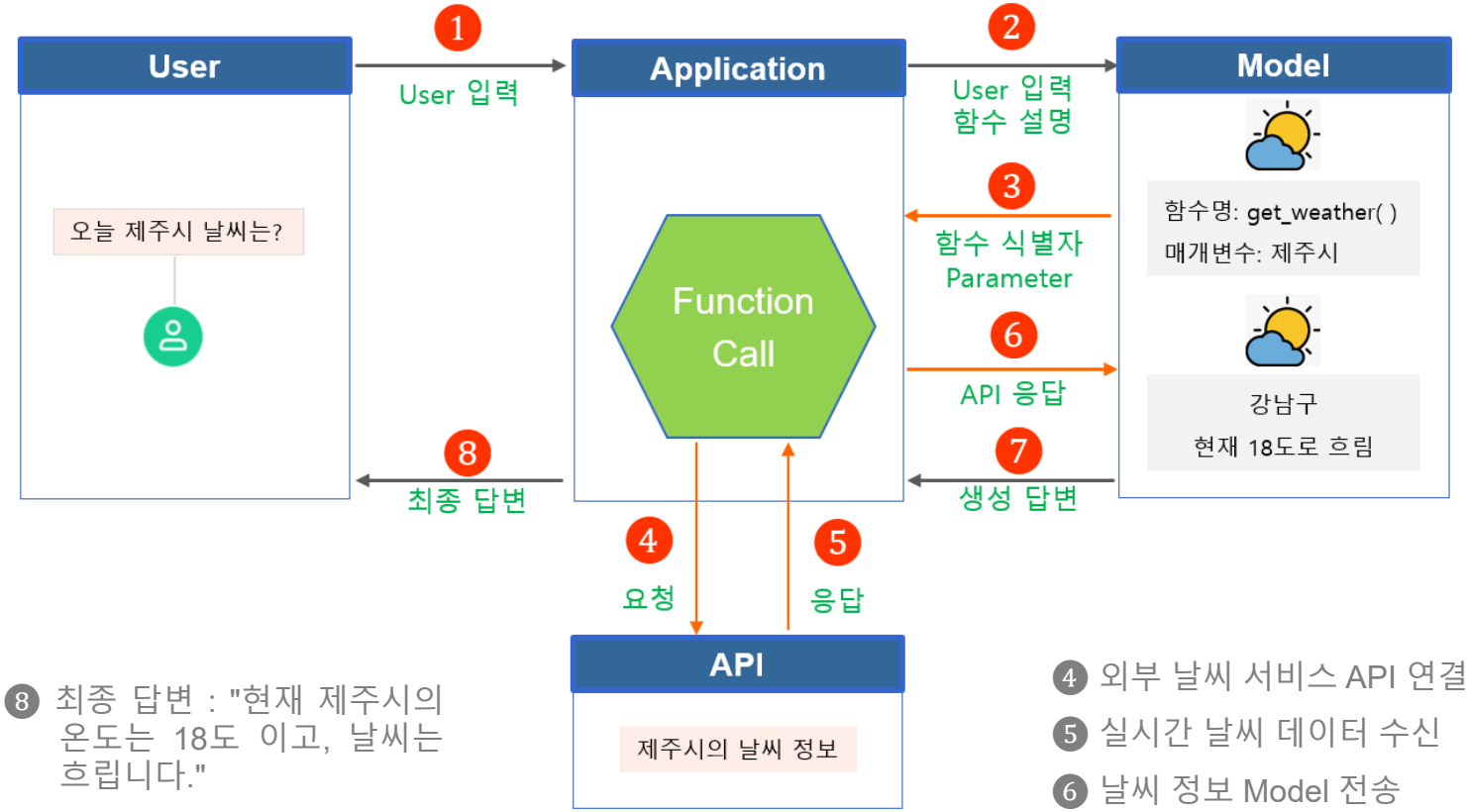
How Function Call Woks? (1)

1. User : "오늘 제주시 날씨는?" 질문
2. Model : 질문 분석
3. Model : 사전에 정의된 함수 중 get_weather() 사용 결정
   - 파라미터 추출 : 위치정보(" 제주시") / Day("2025-11-11")



8. 최종 답변 : "현재 제주시의 온도는 18도 이고, 날씨는 흐립니다."

4. 외부 날씨 서비스 API 연결
5. 실시간 날씨 데이터 수신
6. 날씨 정보 Model 전송
7. Model : 최종 답변 완성

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< Reference: https://www.ncloud-forums.com/topic/475/ >

Case I : Shopping LLM Agent

- Shopping LLM Agent
  - ❖ LLM agent that integrates with external systems
  - ❖ Check the inventory of a specific product
  - ❖ Predefined function
    - ◆ Inventory check: getCount( )
    - ◆ product information check
    - ◆ payment processing

< Reference: https://www.ncloud-forums.com/topic/475/ >

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Case II : Task Automation Agent

- Task Automation Agent
  - ❖ Summarize the latest articles every morning and upload them to the company bulletin board.
  - ❖ Predefined function
    - ◆ Uploadnews( )

< Reference: https://www.ncloud-forums.com/topic/475/ >

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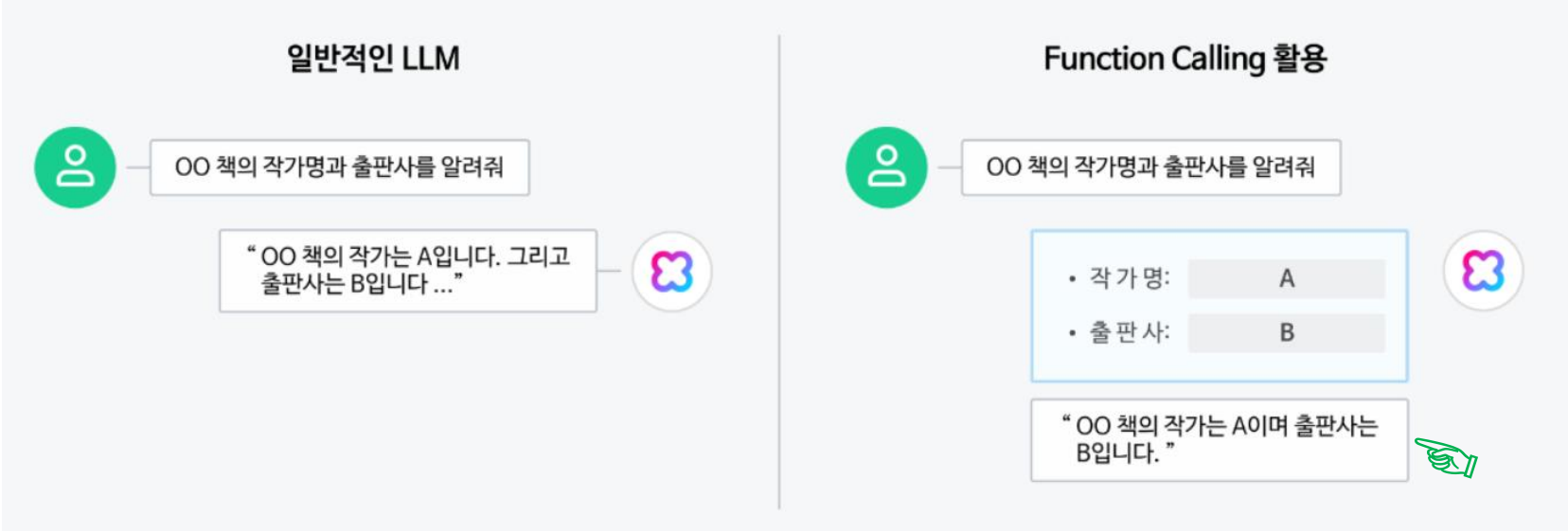
Case III : Extract Information Agent (1)

- Extract Information Agent
  - To extract core data according to a clearly defined format (schema)
  - Sample
    - Extract the author and publisher for a specific book



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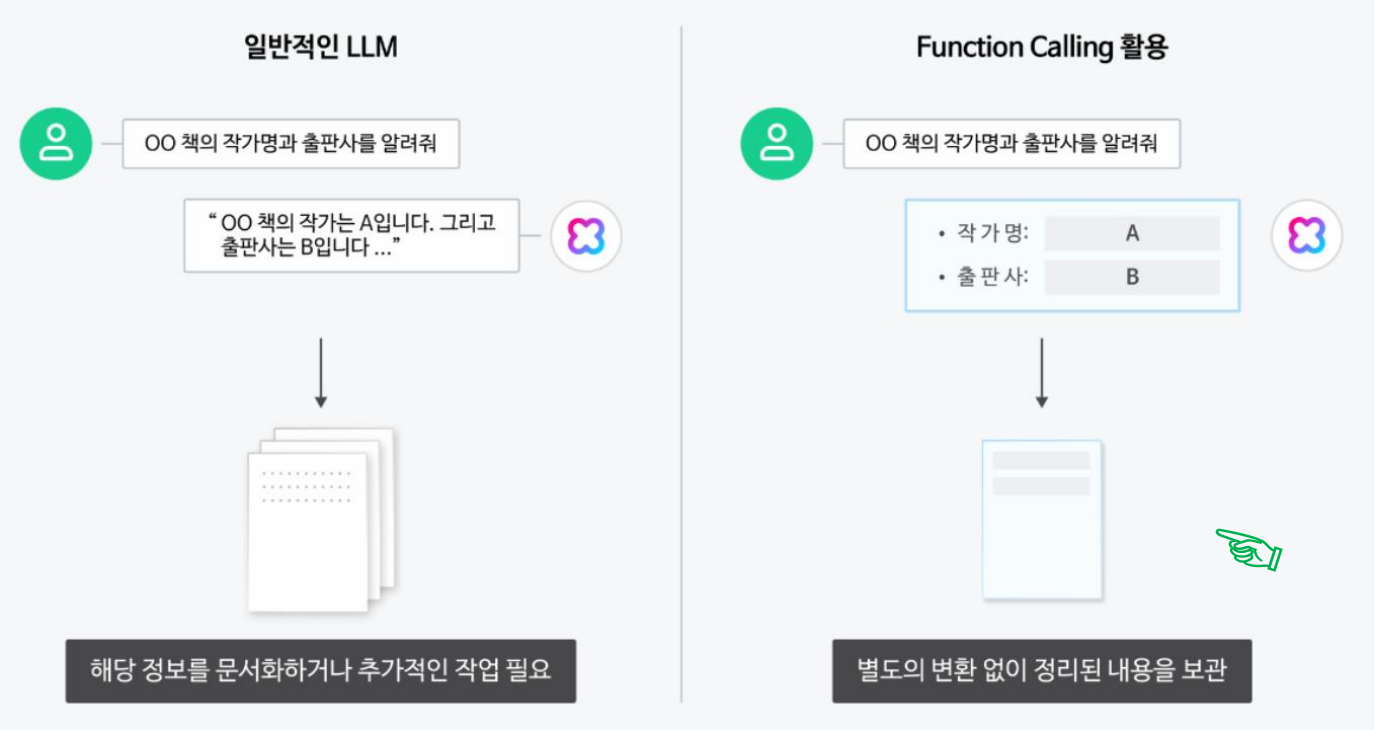
< Reference: https://www.ncloud-forums.com/topic/475/ >

Case III : Extract Information Agent (2)

- Processing result data in JSON format
  - Function calling
    - The extracted data is immediately received and stored in JSON format.



< Reference: https://www.ncloud-forums.com/topic/475/ >

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Practice I : Function Call (1)

Practice I : Function Call (Time : 20minutes)

- Implement a function call that returns the current time
  - File Name : **GPT_ FunctionCall.py**

```
(venvPython312x) PS C:\LLM_Project>
(venvPython312x) PS C:\LLM_Project> & C:/LLM_Project/venvPython312x/Scripts/python.exe c
User     : Hi
ChatCompletionMessage(content='Hello! How can I assist you today?', refusal=None, role='
calls=None, annotations=[])
GPT      : Hello! How can I assist you today?
User     : 지금 몇시인지 알려줘?
ChatCompletionMessage(content=None, refusal=None, role='assistant', audio=None, function
Call(id='call_tvpYQA8HVq4KO9BwHAjONMVy', function=Function(arguments='{}', name='get_cur
2025-11-16 03:10:56
GPT      : 현재 시간은 2025년 11월 16일 오전 3시 10분입니다. 다른 도움이 필요하신가요?
User     : exit
(venvPython312x) PS C:\LLM_Project>
```

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Practice I : Function Call (2)

❖ GPT_Date_Function.py

2. Run

E: > LECTURE > [AI_Agent_프로그래밍] > Source > LLM_AI에이전트 > chap07 > sec01 > GPT_Date_Function.py > ...

```python
from datetime import datetime

def get_current_time():
    now = datetime.now().strftime("%Y-%m-%d %H:%M:%S")
    print(now)
    return now


if __name__ == '__main__':
    get_current_time()
```

1. Coding

```
(venvPython312x) PS C:\LLM_Project> & C:/LLM_Project/venvPython312x/Scripts/python.exe e:/LECTURE/[AI_Agent_프로그래밍]/Source/LLM_AI에이전트/chap07/sec01/GPT_Date_Function.py
2025-11-15 17:19:35
(venvPython312x) PS C:\LLM
```

3. Output

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Practice I : Function Call (3)

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GPT_Date_Function.py

E: > LECTURE > [AI_Agent_프로그래밍] > Source > LLM_AI에이전트 > chap07 > sec01 > GPT_Date_Function.py > ...

```python
from datetime import datetime

def get_current_time():
    now = datetime.now().strftime("%Y-%m-%d %H:%M:%S")
    print(now)
    return now

tools = [          # Displaying as function tool for AI model
    {
        "type": "function",
        "function": {
            "name": "get_current_time",
            "description": "현재 날짜와 시간을 반환합니다.",
        }
    },
]


if __name__ == '__main__':
    get_current_time()

```

5. Run

4. Add Code

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
(venvPython312x) PS C:\LLM_Project> & C:/LLM_Project/venvPython312x/Scripts/python.exe e:/LECTURE/[AI_Agent_프로그래밍]/Source/LLM_AI에이전트/chap07/sec01/GPT_Date_Function.py
2025-11-15 17:19:35
(venvPython312x) PS C:\LLM_Pr
```

6. Output

Practice I : Function Call (4)

❖ GPT_FunctionCall.py

7. Coding

```python
from GPT_Date_Function import get_current_time, tools    # GPT_Date_Function.py
from openai import OpenAI
from dotenv import load_dotenv
import os

load_dotenv()
strOpenAI_API_Key = os.getenv('OPENAI_API_KEY')  # Get API key from .env file

# Call OpenAI() method
client = OpenAI(api_key = strOpenAI_API_Key)

# Function to get AI response
def get_AI_response(strMessages, strTools=None):
    response = client.chat.completions.create(
        model="gpt-4o",
        messages=strMessages,
        tools=strTools,                 # Pass available tools
    )
    return response

messages = [                            # Starting system message
    {"role": "system", "content": "너는 User를 도와주는 상담사야."},
]

while True:...
```

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Practice I : Function Call (5)

8. Coding

GPT_FunctionCall.py

```python
while True:
    user_input = input("User\t: ")  # User input
    if user_input == "exit":        # User wants to exit
        break

    # User message added to conversation history
    messages.append({"role": "user", "content": user_input})

    AI_response = get_AI_response(messages, tools)
    AI_message = AI_response.choices[0].message
    # print(AI_message)                          # AI response for debugging

    tool_calls = AI_message.tool_calls          # AI Response tool_calls
    if tool_calls:                              # If tool_calls exist
        tool_name = tool_calls[0].function.name # Extract function name
        tool_call_id = tool_calls[0].id         # Tool_call ID

        if tool_name == "get_current_time":     # tool_name="get_current_time"
            messages.append({
                "role": "function",             # Set role to "function"
                "tool_call_id": tool_call_id,
                "name": tool_name,
                "content": get_current_time(),  # content=get_current_time()
            })

        AI_response = get_AI_response(messages, tools)
        AI_message = AI_response.choices[0].message
```

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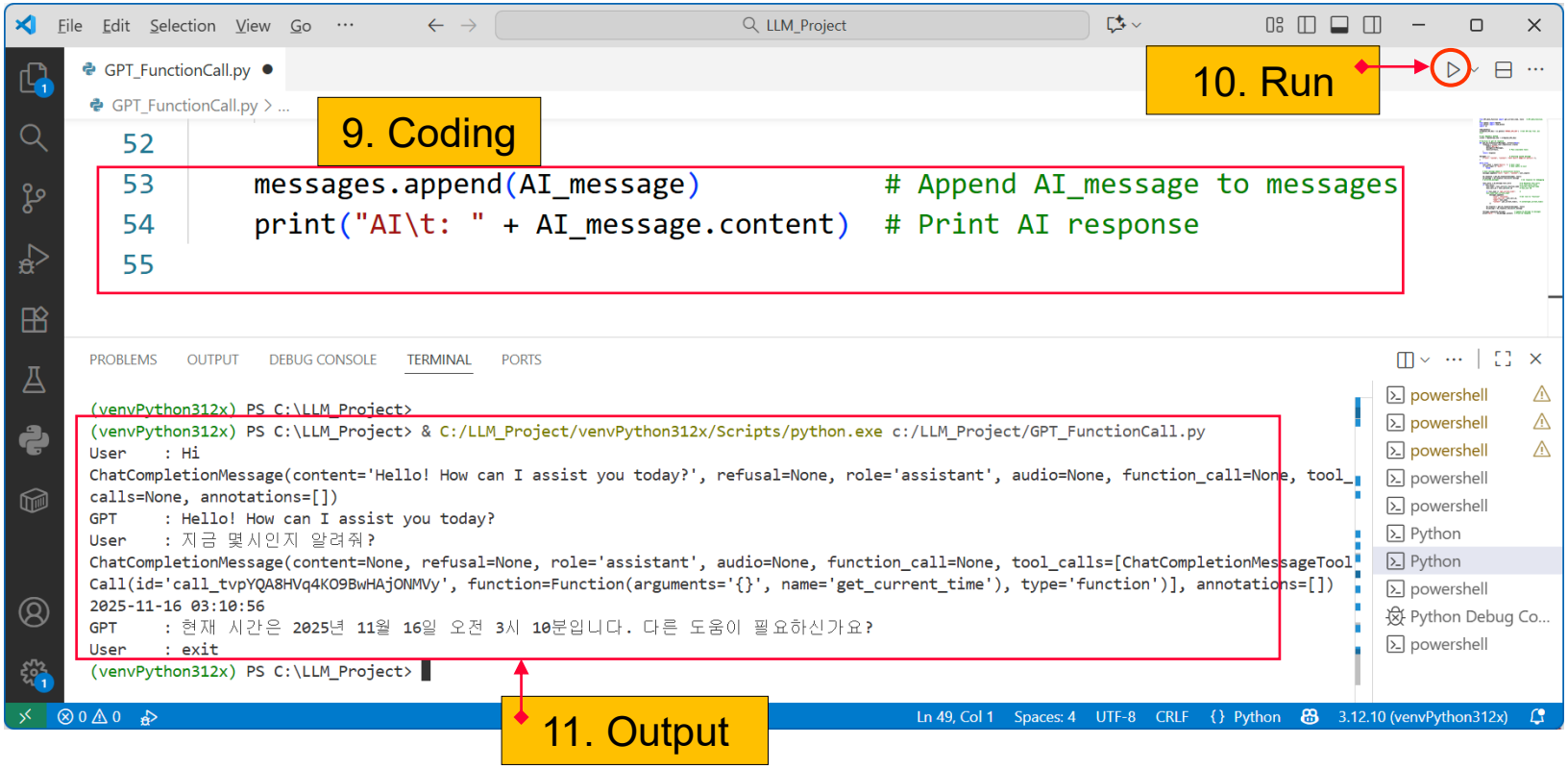
Practice I : Function Call (6)

❖ Run



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Practice II : Timezone Function Call (1)

- Practice II : Timezone Function Call (Time : 20minutes)
  - Implement a function call that returns the time in a specified city.
    - File Name : **GPT_Timezone_FunctionCall.py**

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS Python +

```
(venvPython312x) PS C:\LLM_Project> & C:/LLM_Project/venvPython312x/Scripts/python.exe
c:/LLM_Project/GPT_Timezone_FunctionCall.py
User    : 현재 시간 알려줘
AI      : 어떤 지역(타임존)의 현재 시간이 궁금한가요? 특정 도시나 지역을 알려주시면 해
당 지역의 현재 시간을 알려드리겠습니다.
User    : 서울
2025-11-16 14:10:16 Asia/Seoul
AI      : 서울의 현재 시간은 2025년 11월 16일 일요일 오후 2시 10분입니다.
User    : 런던은?
2025-11-16 05:10:35 Europe/London
AI      : 런던의 현재 시간은 2025년 11월 16일 일요일 오전 5시 10분입니다.
User    : 비슈케크는?
2025-11-16 11:10:48 Asia/Bishkek
AI      : 비슈케크의 현재 시간은 2025년 11월 16일 일요일 오전 11시 10분입니다.
User    : exit
(venvPython312x) PS C:\LLM_Project>
```

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Practice II : Timezone Function Call (2)

❖ GPT_Timezone_Function.py

GPT_Timezone_Function.py

```python
from datetime import datetime
import pytz                          # Importing pytz for timezone handling

def get_current_time(timezone: str = 'Asia/Seoul'):
    tz = pytz.timezone(timezone)                        # Set timezone
    now = datetime.now(tz).strftime("%Y-%m-%d %H:%M:%S")
    now_timezone = f'{now} {timezone}'
    print(now_timezone)
    return now_timezone

tools = [ ...

if __name__ == '__main__':
    get_current_time('America/New_York') # Call get_current_time()
```

1. Coding

```
PS C:\LLM_Project> & C:/LLM_Project/venvPython312x/Scripts/Activate.ps1
(venvPython312x) PS C:\LLM_Project> & C:/LLM_Project/venvPython312x/Scripts/python.exe c:/LLM_Project/GPT_CityDate_Function.py
2025-11-15 23:43:18 America/New_York
(venvPython312x) PS C:\LLM_Project>
```

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Practice II : Timezone Function Call (3)

❖ GPT_Timezone_Function.py : tools

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GPT_Timezone_Function.py

2. Coding

3. Run

```python
tools = [
    {
        "type": "function",
        "function": {
            "name": "get_current_time",
            "description": "해당 타임존의 날짜와 시간을 반환합니다.",
            "parameters": {
                "type": "object",
                "properties": {
                    'timezone': {
                        'type': 'string',
                        'description': '현재 날짜와 시간을 반환할 타임존(예: Asia/Seoul)을 입력하세요.',
                    },
                },
                "required": ['timezone'],
            },
        }
    },
]
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\LLM_Project> & C:/LLM_Project/venvPython312x/Scripts/Activate.ps1
(venvPython312x) PS C:\LLM_Project> & C:/LLM_Project/venvPython312x/Scripts/python.exe c:/LLM_Project/GPT_CityDate_Function.py
2025-11-15 23:43:18 America/New_York
(venvPython312x) PS C:\LLM_Project>
```

4. Output

Practice II : Timezone Function Call (4)

❖ GPT_Timezone_FunctionCall.py

5. Coding

GPT_Timezone_FunctionCall.py

```python
from GPT_CityDate_Function import get_current_time, tools   # GPT_Date_Function.py
from openai import OpenAI
from dotenv import load_dotenv
import os
import json

load_dotenv()
strOpenAI_API_Key = os.getenv('OPENAI_API_KEY')  # Get API key from .env file

# Call OpenAI() method
client = OpenAI(api_key = strOpenAI_API_Key)

# Function to get AI response
def get_AI_response(strMessages, strTools=None):...

messages = [                          # Starting system message...

while True:...
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
(venvPython312x) PS C:\LLM_Project> & C:/LLM_Project/venvPython312x/Scripts/python.exe c:/LLM_Project/GPT_CityDate_Function.py
2025-11-15 23:43:18 America/New_York
(venvPython312x) PS C:\LLM_Project> & C:/LLM_Project/venvPython312x/Scripts/python.exe c:/LLM_Project/GPT_Timezone_FunctionCall.py
User    : 현재 시간 알려줘
AI      : 어떤 지역(타임존)의 현재 시간이 궁금한가요? 특정 도시나 지역을 알려주시면 해당 지역의 현재 시간을 알려드리겠습니다.
User    : 서울
2025-11-16 14:10:16 Asia/Seoul
AI      : 서울의 현재 시간은 2025년 11월 16일 일요일 오후 2시 10분입니다.
User    : 런던은?
```

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Practice II : Timezone Function Call (5)

❖ GPT_Timezone_FunctionCall.py : **While True**

```python
while True:
    user_input = input("User\t: ")  # User input
    if user_input == "exit":        # User wants to exit
        break

    # User message added to conversation history
    messages.append({"role": "user", "content": user_input})

    AI_response = get_AI_response(messages, tools)
    AI_message = AI_response.choices[0].message
    # print(AI_message)                          # AI response for debugging

    tool_calls = AI_message.tool_calls           # AI Response tool_calls
    if tool_calls:                               # If tool_calls exist
        tool_name = tool_calls[0].function.name # Extract function name
        tool_call_id = tool_calls[0].id          # Tool_call ID

        # String -> Dictionary
        arguments = json.loads(tool_calls[0].function.arguments)

        if tool_name == "get_current_time":      # tool_name="get_current_time"
            messages.append({
                "role": "function",              # Set role to "function"
                "tool_call_id": tool_call_id,
                "name": tool_name,
                # content=get_current_time() with timezone
                "content": get_current_time(timezone=arguments['timezone']),
            })
```

6. Coding

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Practice II : Timezone Function Call (6)

❖ Run

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```
(venvPython312x) PS C:\LLM_Project> & C:/LLM_Project/venvPython312x/Scripts/python.exe
c:/LLM_Project/GPT_Timezone_FunctionCall.py
User    : 현재 시간 알려줘
AI      : 어떤 지역(타임존)의 현재 시간이 궁금한가요? 특정 도시나 지역을 알려주시면 해
당 지역의 현재 시간을 알려드리겠습니다.
User    : 서울
2025-11-16 14:10:16 Asia/Seoul
AI      : 서울의 현재 시간은 2025년 11월 16일 일요일 오후 2시 10분입니다.
User    : 런던은?
2025-11-16 05:10:35 Europe/London
AI      : 런던의 현재 시간은 2025년 11월 16일 일요일 오전 5시 10분입니다.
User    : 비슈케크는?
2025-11-16 11:10:48 Asia/Bishkek
AI      : 비슈케크의 현재 시간은 2025년 11월 16일 일요일 오전 11시 10분입니다.
User    : exit
(venvPython312x) PS C:\LLM_Project>
```

7. Output

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Practice III : Multi-Timezone Function Call (1)

Practice III : Multi-Timezone Function Call (Time : 20minutes)

- Implement a function call that returns the time in a specified multiple cities.
  - File Name : GPT_MultiTimezone_FunctionCall.py

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS Python

```
(venvPython312x) PS C:\LLM_Project>
(venvPython312x) PS C:\LLM_Project> & C:/LLM_Project/venvPython312x/Scripts/python.exe
c:/LLM_Project/GPT_MultiTimezone_FunctionCall.py
User    : 서울, 런던, 비슈케크는 몇시야?
2025-11-16 15:03:03 Asia/Seoul
2025-11-16 06:03:03 Europe/London
2025-11-16 12:03:03 Asia/Bishkek
AI      : 현재 시간은 다음과 같습니다:

- 서울: 2025년 11월 16일 오후 3시 3분
- 런던: 2025년 11월 16일 오전 6시 3분
- 비슈케크: 2025년 11월 16일 오후 12시 3분
User    : exit
(venvPython312x) PS C:\LLM_Project>
```

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Practice III : Multi-Timezone Function Call (2)

❖ GPT_MultiTimezone_FunctionCall.py

```python
while True:
    user_input = input("User\t: ")  # User input
    if user_input == "exit":        # User wants to exit
        break

    # User message added to conversation history
    messages.append({"role": "user", "content": user_input})

    AI_response = get_AI_response(messages, tools)
    AI_message = AI_response.choices[0].message
    # print(AI_message)                        # AI response for debugging

    tool_calls = AI_message.tool_calls         # AI Response tool_calls
    if tool_calls:                             # If tool_calls exist...

    messages.append(AI_message)            # Append AI_message to messages
    print("AI\t: " + AI_message.content)   # Print AI response
```

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Practice III : Multi-Timezone Function Call (3)

❖ GPT_MultiTimezone_FunctionCall.py : **if tool_calls**

GPT_MultiTimezone_FunctionCall.py

```python
    tool_calls = AI_message.tool_calls          # AI Response tool_calls
    if tool_calls:                              # If tool_calls exist
        for tool_call in tool_calls:
            tool_name = tool_call.function.name # Extract function name
            tool_call_id = tool_call.id         # Tool_call ID

            # String -> Dictionary
            arguments = json.loads(tool_call.function.arguments)

            if tool_name == "get_current_time":     # tool_name="get_current_time"
                messages.append({
                    "role": "function",             # Set role to "function"
                    "tool_call_id": tool_call_id,
                    "name": tool_name,
                    # content=get_current_time() with timezone
                    "content": get_current_time(timezone=arguments['timezone']),
                })
        # Function execution complete message
        messages.append({"role": "system", "content": "이제 주어진 결과를 바탕으로 답변한다."})

        AI_response = get_AI_response(messages, tools)
        AI_message = AI_response.choices[0].message

    messages.append(AI_message)          # Append AI_message to messages
    print("AI\t: " + AI_message.content) # Print AI response
```

Coding

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Practice III : Multi-Timezone Function Call (6)

❖ Run

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS Python

```
(venvPython312x) PS C:\LLM_Project>
(venvPython312x) PS C:\LLM_Project> & C:/LLM_Project/venvPython312x/Scripts/python.exe
c:/LLM_Project/GPT_MultiTimezone_FunctionCall.py
User     : 서울, 런던, 비슈케크는 몇시야?
2025-11-16 15:03:03 Asia/Seoul
2025-11-16 06:03:03 Europe/London
2025-11-16 12:03:03 Asia/Bishkek
AI       : 현재 시간은 다음과 같습니다:

- 서울: 2025년 11월 16일 오후 3시 3분
- 런던: 2025년 11월 16일 오전 6시 3분
- 비슈케크: 2025년 11월 16일 오후 12시 3분
User     : exit
(venvPython312x) PS C:\LLM_Project>
```

Output

LangChain

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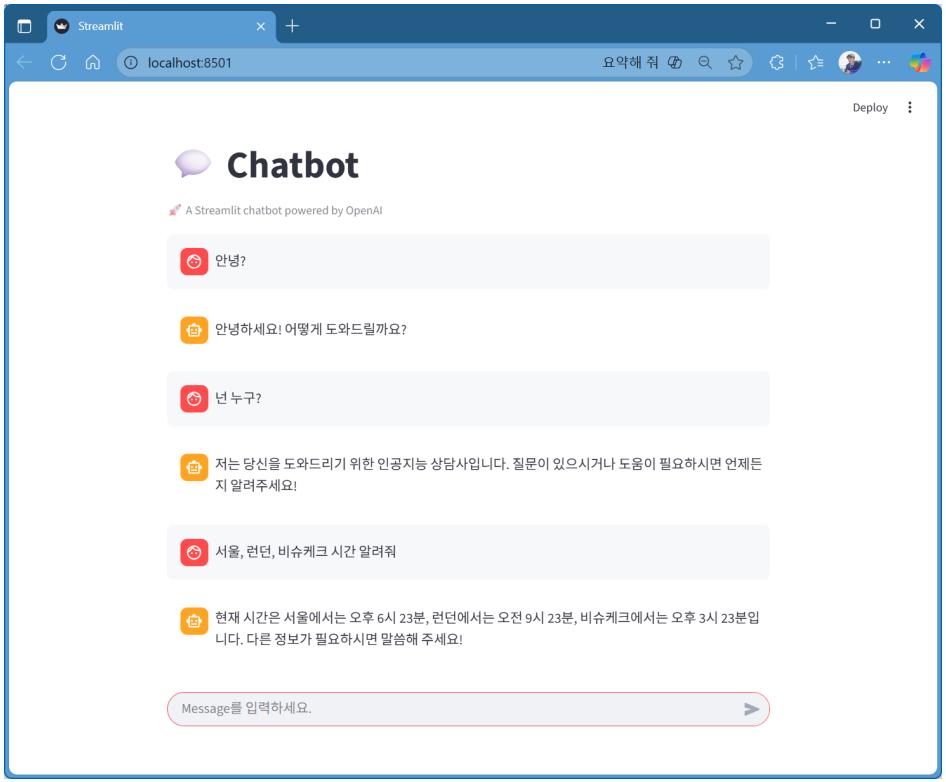
LangGraph

Practice IV : Function Call in Streamlit (1)

- Practice IV : Function Call in Streamlit (Time : 20minutes)
  - Implement a function call in Streamlit that returns the time in several specified cities.
    - File Name : GPT_Streamlit_FunctionCall.py
      - messages ▶ st.session_state.messages



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Practice IV : Function Call in Streamlit (2)

❖ GPT_Streamlit_FunctionCall.py

GPT_Streamlit_FunctionCall.py

```python
from GPT_CityDate_Function import get_current_time, tools   # GPT_Date_Function.py
from openai import OpenAI
from dotenv import load_dotenv
import os
import json
import streamlit as st

load_dotenv()

# Input api_key in the sidebar
with st.sidebar:
    strOpenAI_API_Key = os.getenv('OPENAI_API_KEY')  # Get API key from .env file

# Call OpenAI() method
client = OpenAI(api_key = strOpenAI_API_Key)

# Function to get AI response
def get_AI_response(strMessages, strTools=None):...

st.title("💬 Chatbot")                    # Streamlit UI
st.caption("🚀 A Streamlit chatbot powered by OpenAI")

if "messages" not in st.session_state:
    st.session_state["messages"] = [   # Initial message of st.session_state
        {"role": "system", "content": "너는 사용자를 도와주는 상담사야."},
    ]
```

1. Coding

LangChain

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Practice IV : Function Call in Streamlit (3)

❖ messages ▶ st.session_state.messages

GPT_Streamlit_FunctionCall.py

2. Coding

```python
for msg in st.session_state.messages:
    if msg["role"] in ["assistant", "user"]:  # Only for assistant or user messages
        st.chat_message(msg["role"]).write(msg["content"])

if user_input := st.chat_input("Message를 입력하세요."):   # Chat User input
    # Append user message to conversation history
    st.session_state.messages.append({"role": "user", "content": user_input})
    st.chat_message("user").write(user_input)  # User message to browser output

    # Show loading spinner
    with st.spinner("AI is thinking..."):
        AI_response = get_AI_response(st.session_state.messages, tools)

    # Validate AI_response structure
    if AI_response.choices and hasattr(AI_response.choices[0], "message"):
        AI_message = AI_response.choices[0].message
    else:
        AI_message = {"content": "No AI response... please try again."}
```

LangChain

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Practice IV : Function Call in Streamlit (4)

LangChain

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GPT_Streamlit_FunctionCall.py

```python
    # Check for tool_calls in AI response
    tool_calls = AI_message.tool_calls          # AI Response tool_calls
    if tool_calls:                              # If tool_calls exist
        for tool_call in tool_calls:
            tool_name = tool_call.function.name # Extract function name
            tool_call_id = tool_call.id         # Tool_call ID

            # String -> Dictionary
            arguments = json.loads(tool_call.function.arguments)

            # Add function execution result to session state
            if tool_name == "get_current_time":     # tool_name="get_current_time"
                st.session_state.messages.append({
                    "role": "function",             # Set role to "function"
                    "tool_call_id": tool_call_id,
                    "name": tool_name,
                    "content": get_current_time(timezone=arguments['timezone']),
                })

        # Add system message after tool execution
        st.session_state.messages.append({"role": "system", "content": "이제 주어진 결과를 바탕으로 답변한다."})

        # Get final AI response after tool execution
        with st.spinner("Organizing answers based on AI results..."):
            AI_response = get_AI_response(st.session_state.messages)
        AI_message = AI_response.choices[0].message
```

3. Coding

Practice IV : Function Call in Streamlit (5)

3. Coding

GPT_Streamlit_FunctionCall.py

```python
# Append AI_message.content to session state
st.session_state.messages.append({"role": "assistant",
                                  "content": AI_message.content })

# Display AI response to browser
st.chat_message("assistant").write(AI_message.content)
```

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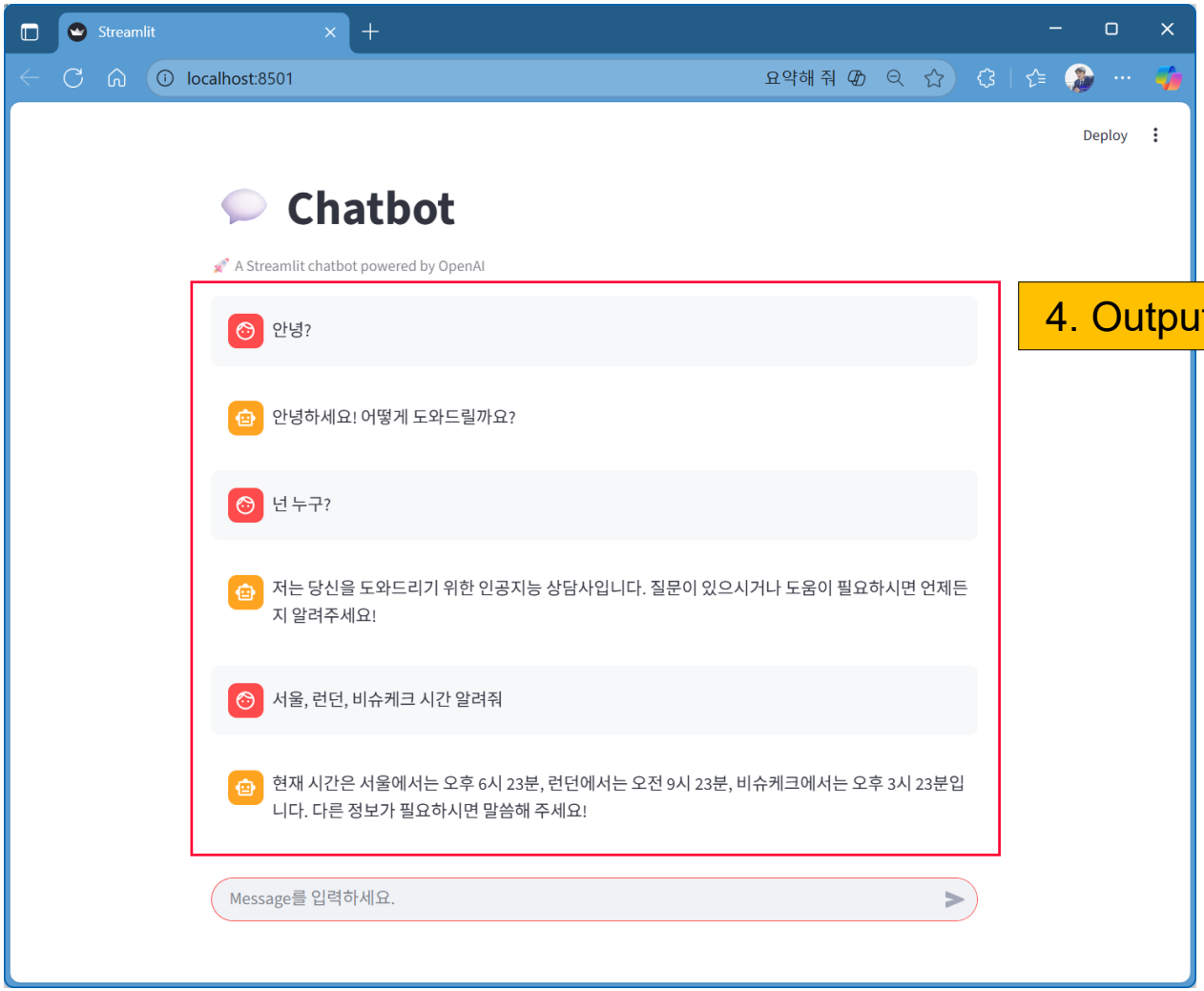
Practice IV : Function Call in Streamlit (6)

❖ Run



4. Output

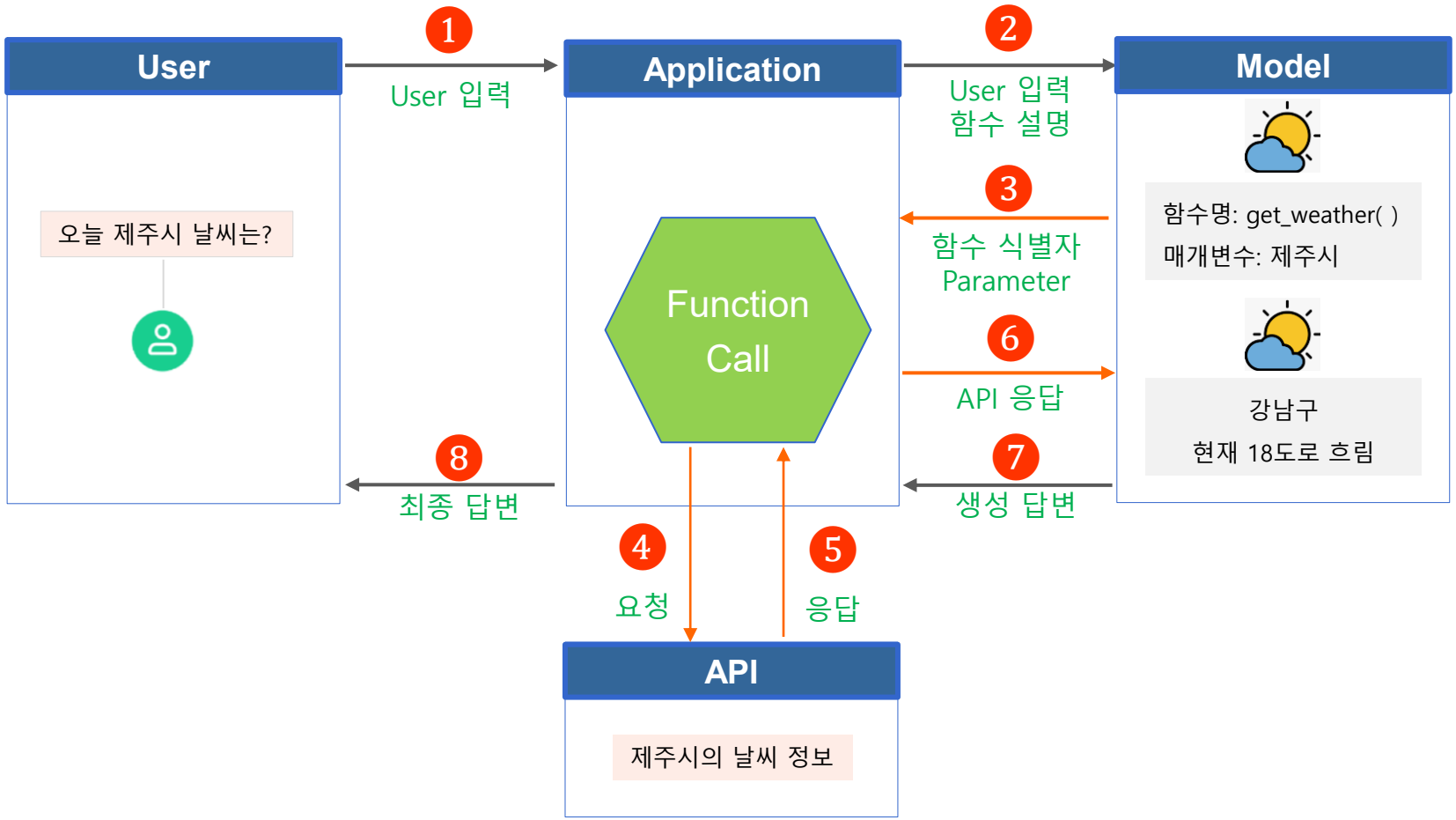
LangChain

OpenAI

LangGraph

Summary

- What is a function calling?
- How Function Call Woks ?
  - Case I : Shopping LLM Agent
  - Case II : Task Automation Agent
  - Case III : Extract Information Agent
  - Practice I : Function Call
  - Practice II : Timezone Function Call
  - Practice III : Multi-Timezone Function Call
  - Practice IV : Function Call in Streamlit



LangChain

OpenAI

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